

KISSsoft - Release 03/2017

Team-SolidSQUAD

Файл

Имя : Reductor
Изменил: Admin дата: 01.03.2026 в: 19:03:45

АНАЛИЗ РЕЗУЛЬТАТОВ(Оценка важных свойств)

Комментарий:

№ = номер варианта
diff_i = отклонение от заданного передаточного отношения в %
кг = вес в кг
слайд = удельное скольжение (максимальное значение)
v.Slide = Скорость скольжения (м/с, максимальное значение)
AC/AE = Рабочий участок линии зацепления на входе AC / рабочий участок линии зацепления AE
(характеристики при трении)
del_cg = Стандартное отклонение жесткости при обкатке (Н/мм/мум)
1-eta = Потери в % (1.0-общий КПД)
Safety = запас прочности (ножка зуба и боковая поверхность, 0 = высоко, 1 = достаточно, 2 = низко)
(SF-min: 0.60/ 1.20/ 1.40 SH-min: 0.60/ 0.90/ 1.00)
Summary = Общая оценка (оценочно)
(50.0%:del_cg 20.0%:diff_i 100.0%:kg 35.0%:Slide 0.0%:v.Slide
0.0%:AC/AE 10.0%:1-eta 100.0%:Safety)

(Для таблицы справедливо: Чем меньше число, тем лучше!)

No.	diff_i	kg	Slide	v.Slide	AC/AE	del_cg	1-eta	Safety	Summary
1	4.762	1.874	0.720	1.192	0.471	0.194	0.791	0.835	0.391
2	-1.149	1.854	0.711	1.213	0.474	0.195	0.762	0.808	0.368
3	1.149	1.868	1.184	1.392	0.560	0.238	0.858	0.798	0.367
4	1.149	1.865	1.025	1.276	0.514	0.233	0.837	0.798	0.366
5	1.149	1.861	0.880	1.314	0.469	0.228	0.838	0.802	0.367
6	3.448	1.876	1.540	1.452	0.551	0.262	0.948	0.806	0.380

7	3.448	1.872	1.309	1.330	0.506	0.259	0.927	0.808	0.380
8	3.448	1.868	1.106	1.408	0.462	0.253	0.931	0.812	0.380
9	-4.444	1.849	1.160	1.436	0.564	0.239	0.832	0.775	0.368
10	-4.444	1.846	1.008	1.318	0.519	0.235	0.809	0.775	0.367
11	-4.444	1.843	0.870	1.336	0.473	0.230	0.809	0.778	0.368
12	-2.222	1.856	1.504	1.499	0.556	0.262	0.919	0.781	0.365
13	-2.222	1.853	1.286	1.376	0.511	0.259	0.896	0.781	0.364
14	-2.222	1.849	1.092	1.430	0.467	0.256	0.898	0.784	0.364
15	0.000	1.863	2.067	1.550	0.548	0.275	1.021	0.808	0.373
16	0.000	1.860	1.713	1.419	0.504	0.280	0.999	0.814	0.373
17	1.149	1.909	0.718	1.216	0.473	0.205	0.762	0.803	0.367
18	3.448	1.924	1.198	1.385	0.558	0.243	0.857	0.793	0.374
19	3.448	1.920	1.035	1.269	0.512	0.238	0.836	0.794	0.373
20	3.448	1.916	0.887	1.316	0.467	0.232	0.838	0.797	0.373
21	-4.444	1.890	0.709	1.237	0.477	0.206	0.735	0.781	0.368
22	-2.222	1.904	1.173	1.428	0.562	0.244	0.830	0.771	0.360
23	-2.222	1.900	1.018	1.311	0.517	0.240	0.809	0.771	0.359
24	-2.222	1.897	0.877	1.337	0.471	0.234	0.809	0.774	0.359
25	0.000	1.911	1.519	1.489	0.554	0.266	0.917	0.777	0.357
26	0.000	1.907	1.295	1.366	0.509	0.263	0.895	0.777	0.356
27	0.000	1.904	1.098	1.429	0.465	0.257	0.898	0.781	0.356
28	-3.226	1.899	2.033	1.588	0.552	0.277	0.989	0.776	0.371
29	3.448	1.965	0.725	1.219	0.471	0.206	0.761	0.798	0.374
30	-2.222	1.945	0.716	1.239	0.475	0.213	0.735	0.776	0.360
31	0.000	1.959	1.184	1.420	0.560	0.247	0.829	0.767	0.352
32	0.000	1.955	1.026	1.302	0.515	0.242	0.807	0.767	0.351
33	0.000	1.952	0.882	1.338	0.469	0.238	0.808	0.770	0.351
34	2.222	1.967	1.533	1.478	0.552	0.268	0.915	0.773	0.364
35	2.222	1.963	1.304	1.355	0.507	0.266	0.894	0.774	0.363
36	2.222	1.959	1.103	1.429	0.463	0.260	0.897	0.777	0.363
37	-3.226	1.947	1.499	1.526	0.557	0.269	0.888	0.751	0.358
38	-3.226	1.944	1.283	1.400	0.512	0.266	0.866	0.751	0.357
39	-3.226	1.940	1.091	1.450	0.467	0.262	0.867	0.754	0.357
40	0.000	2.001	0.722	1.242	0.473	0.214	0.734	0.772	0.352
41	2.222	2.016	1.197	1.413	0.558	0.251	0.828	0.763	0.358
42	2.222	2.012	1.035	1.295	0.513	0.247	0.807	0.762	0.357
43	2.222	2.008	0.889	1.339	0.468	0.242	0.809	0.766	0.358

44	4.444	2.024	1.549	1.468	0.550	0.271	0.913	0.769	0.371
45	4.444	2.020	1.314	1.345	0.506	0.268	0.893	0.770	0.370
46	4.444	2.016	1.109	1.428	0.461	0.262	0.897	0.774	0.370
47	-3.226	1.996	1.173	1.456	0.563	0.251	0.803	0.743	0.353
48	-3.226	1.992	1.019	1.337	0.517	0.249	0.782	0.742	0.352
49	-3.226	1.989	0.878	1.360	0.472	0.244	0.782	0.745	0.352
50	-1.075	2.003	1.513	1.515	0.555	0.273	0.886	0.747	0.350
51	-1.075	1.999	1.292	1.390	0.510	0.268	0.865	0.748	0.349
52	-1.075	1.996	1.096	1.449	0.466	0.264	0.867	0.751	0.349
53	2.222	2.062	0.841	1.226	0.519	0.227	0.734	0.765	0.358
54	2.222	2.058	0.728	1.245	0.472	0.222	0.734	0.768	0.359
55	4.444	2.074	1.209	1.405	0.556	0.253	0.827	0.759	0.365
56	4.444	2.070	1.044	1.288	0.511	0.249	0.807	0.759	0.364
57	4.444	2.066	0.894	1.340	0.466	0.245	0.809	0.762	0.365
58	-3.226	2.041	0.829	1.264	0.522	0.229	0.711	0.746	0.353
59	-3.226	2.038	0.720	1.265	0.476	0.223	0.709	0.749	0.354
60	-1.075	2.052	1.184	1.448	0.561	0.254	0.802	0.739	0.346
61	-1.075	2.049	1.027	1.328	0.516	0.251	0.781	0.739	0.344
62	-1.075	2.045	0.884	1.360	0.470	0.247	0.781	0.742	0.345
63	1.075	2.060	1.528	1.505	0.553	0.275	0.885	0.743	0.349
64	1.075	2.056	1.301	1.379	0.509	0.272	0.864	0.742	0.348
65	1.075	2.052	1.101	1.449	0.464	0.267	0.867	0.746	0.348
66	-4.167	2.040	1.495	1.552	0.558	0.276	0.860	0.719	0.349
67	-4.167	2.037	1.280	1.424	0.513	0.271	0.838	0.717	0.347
68	-4.167	2.034	1.089	1.470	0.468	0.270	0.839	0.720	0.347
69	4.444	2.120	0.850	1.222	0.517	0.234	0.734	0.761	0.365
70	4.444	2.116	0.735	1.247	0.470	0.224	0.734	0.764	0.365
71	-1.075	2.098	0.837	1.259	0.521	0.235	0.710	0.742	0.345
72	-1.075	2.095	0.726	1.267	0.474	0.225	0.709	0.745	0.346
73	1.075	2.110	1.195	1.440	0.559	0.257	0.801	0.736	0.345
74	1.075	2.106	1.035	1.320	0.514	0.254	0.780	0.735	0.344
75	1.075	2.102	0.889	1.361	0.468	0.249	0.781	0.738	0.344
76	3.226	2.118	1.542	1.494	0.551	0.275	0.883	0.739	0.355
77	3.226	2.114	1.310	1.369	0.507	0.274	0.863	0.737	0.353
78	3.226	2.110	1.106	1.448	0.462	0.270	0.866	0.742	0.354
79	-4.167	2.090	1.173	1.483	0.563	0.258	0.778	0.718	0.348
80	-4.167	2.086	1.019	1.361	0.518	0.256	0.757	0.717	0.346

81	-4.167	2.083	0.879	1.382	0.473	0.251	0.756	0.720	0.347
82	-2.083	2.097	1.508	1.540	0.556	0.277	0.858	0.714	0.341
83	-2.083	2.094	1.288	1.413	0.511	0.275	0.837	0.713	0.339
84	-2.083	2.090	1.094	1.469	0.466	0.272	0.838	0.716	0.339
85	1.075	2.157	0.844	1.253	0.519	0.236	0.709	0.739	0.345
86	1.075	2.153	0.732	1.269	0.473	0.230	0.709	0.742	0.345
87	3.226	2.169	1.208	1.432	0.557	0.260	0.800	0.732	0.352
88	3.226	2.165	1.044	1.313	0.512	0.257	0.780	0.732	0.350
89	3.226	2.161	0.895	1.362	0.467	0.252	0.782	0.735	0.351
90	-4.167	2.136	0.832	1.290	0.523	0.237	0.688	0.722	0.348
91	-4.167	2.133	0.723	1.290	0.476	0.232	0.686	0.724	0.348
92	-2.083	2.148	1.184	1.475	0.561	0.261	0.777	0.715	0.340
93	-2.083	2.144	1.027	1.353	0.516	0.258	0.756	0.714	0.339
94	-2.083	2.140	0.885	1.383	0.471	0.254	0.756	0.717	0.339
95	0.000	2.155	1.522	1.530	0.554	0.279	0.857	0.710	0.334
96	0.000	2.152	1.297	1.403	0.510	0.276	0.836	0.710	0.332
97	0.000	2.148	1.099	1.468	0.465	0.273	0.838	0.713	0.332
98	3.226	2.221	0.976	1.362	0.564	0.249	0.729	0.734	0.352
99	3.226	2.216	0.852	1.248	0.517	0.243	0.709	0.735	0.351
100	3.226	2.212	0.737	1.271	0.471	0.234	0.709	0.738	0.352
101	-2.083	2.199	0.959	1.401	0.567	0.250	0.707	0.718	0.341
102	-2.083	2.195	0.839	1.284	0.521	0.244	0.687	0.718	0.340
103	-2.083	2.191	0.729	1.291	0.475	0.239	0.686	0.721	0.341
104	0.000	2.207	1.195	1.466	0.560	0.263	0.776	0.711	0.333
105	0.000	2.203	1.035	1.345	0.514	0.261	0.755	0.710	0.332
106	0.000	2.199	0.890	1.383	0.469	0.257	0.756	0.713	0.332
107	2.083	2.215	1.535	1.520	0.553	0.280	0.855	0.706	0.340
108	2.083	2.211	1.305	1.393	0.508	0.279	0.834	0.706	0.338
109	2.083	2.207	1.103	1.467	0.463	0.275	0.837	0.710	0.339
110	-3.030	2.194	1.503	1.565	0.557	0.282	0.832	0.686	0.334
111	-3.030	2.190	1.285	1.436	0.512	0.278	0.810	0.687	0.333
112	-3.030	2.187	1.091	1.487	0.467	0.278	0.811	0.690	0.333
113	0.000	2.258	0.969	1.395	0.566	0.252	0.707	0.714	0.333
114	0.000	2.254	0.847	1.279	0.519	0.247	0.687	0.715	0.333
115	0.000	2.250	0.735	1.293	0.473	0.240	0.686	0.718	0.333
116	2.083	2.267	1.206	1.458	0.558	0.264	0.775	0.708	0.339
117	2.083	2.263	1.043	1.337	0.513	0.263	0.755	0.707	0.338

118	2.083	2.258	0.895	1.383	0.468	0.257	0.756	0.710	0.338
119	4.167	2.275	1.548	1.509	0.551	0.280	0.853	0.703	0.346
120	4.167	2.271	1.313	1.382	0.506	0.280	0.833	0.704	0.345
121	4.167	2.266	1.107	1.465	0.461	0.277	0.837	0.707	0.345
122	-3.030	2.245	1.183	1.500	0.562	0.265	0.753	0.688	0.334
123	-3.030	2.242	1.027	1.377	0.517	0.263	0.733	0.680	0.330
124	-3.030	2.238	0.885	1.404	0.472	0.259	0.732	0.685	0.331
125	-1.010	2.253	1.515	1.554	0.556	0.280	0.829	0.683	0.327
126	-1.010	2.249	1.292	1.425	0.511	0.281	0.808	0.684	0.326
127	-1.010	2.245	1.095	1.485	0.466	0.280	0.810	0.687	0.326
128	2.083	2.319	0.978	1.390	0.564	0.257	0.706	0.711	0.340
129	2.083	2.315	0.855	1.274	0.518	0.251	0.687	0.712	0.339
130	2.083	2.311	0.740	1.295	0.472	0.246	0.686	0.715	0.340
131	4.167	2.328	1.217	1.451	0.556	0.269	0.774	0.705	0.346
132	4.167	2.324	1.051	1.330	0.511	0.266	0.755	0.704	0.345
133	4.167	2.319	0.900	1.384	0.466	0.261	0.756	0.707	0.345
134	-3.030	2.297	0.961	1.428	0.567	0.258	0.686	0.696	0.336
135	-3.030	2.294	0.842	1.310	0.521	0.253	0.666	0.696	0.336
136	-3.030	2.290	0.732	1.315	0.475	0.247	0.665	0.699	0.336
137	-1.010	2.305	1.193	1.492	0.560	0.270	0.752	0.680	0.325
138	-1.010	2.301	1.035	1.369	0.515	0.268	0.732	0.676	0.322
139	-1.010	2.298	0.890	1.404	0.470	0.263	0.732	0.680	0.323
140	1.010	2.314	1.528	1.544	0.554	0.282	0.828	0.680	0.327
141	1.010	2.309	1.300	1.415	0.509	0.281	0.807	0.681	0.326
142	1.010	2.305	1.100	1.484	0.464	0.281	0.810	0.684	0.326
143	-3.922	2.293	1.498	1.589	0.558	0.284	0.806	0.663	0.329
144	-3.922	2.289	1.281	1.458	0.513	0.281	0.785	0.663	0.328
145	-3.922	2.285	1.089	1.505	0.468	0.283	0.785	0.666	0.328
146	4.167	2.382	0.988	1.384	0.562	0.259	0.705	0.708	0.346
147	4.167	2.377	0.862	1.269	0.516	0.254	0.686	0.709	0.346
148	4.167	2.373	0.746	1.297	0.470	0.248	0.687	0.712	0.346
149	-1.010	2.358	0.970	1.422	0.566	0.260	0.685	0.693	0.329
150	-1.010	2.354	0.849	1.304	0.520	0.255	0.666	0.693	0.329
151	-1.010	2.350	0.737	1.317	0.474	0.249	0.665	0.696	0.329
152	1.010	2.367	1.204	1.484	0.559	0.272	0.751	0.675	0.324
153	1.010	2.362	1.042	1.361	0.514	0.267	0.731	0.672	0.321
154	1.010	2.358	0.895	1.403	0.468	0.266	0.732	0.676	0.322

155	3.030	2.375	1.540	1.533	0.552	0.282	0.826	0.678	0.333
156	3.030	2.371	1.307	1.404	0.507	0.284	0.806	0.678	0.332
157	3.030	2.366	1.103	1.482	0.462	0.280	0.809	0.682	0.332
158	-3.922	2.345	1.182	1.525	0.563	0.273	0.731	0.654	0.324
159	-3.922	2.342	1.027	1.401	0.518	0.270	0.711	0.653	0.323
160	-3.922	2.338	0.886	1.424	0.472	0.268	0.710	0.657	0.324
161	-1.961	2.353	1.508	1.577	0.557	0.285	0.804	0.660	0.322
162	-1.961	2.349	1.287	1.446	0.512	0.284	0.783	0.661	0.321
163	-1.961	2.345	1.091	1.502	0.466	0.282	0.784	0.663	0.321
164	1.010	2.420	0.979	1.416	0.564	0.263	0.684	0.690	0.329
165	1.010	2.416	0.856	1.298	0.518	0.259	0.665	0.690	0.328
166	1.010	2.412	0.742	1.318	0.472	0.254	0.665	0.693	0.329
167	3.030	2.429	1.214	1.476	0.557	0.275	0.750	0.671	0.330
168	3.030	2.425	1.049	1.353	0.512	0.272	0.731	0.668	0.328
169	3.030	2.420	0.900	1.403	0.467	0.267	0.732	0.672	0.328
170	-3.922	2.398	0.963	1.454	0.568	0.264	0.666	0.662	0.327
171	-3.922	2.395	0.844	1.335	0.522	0.260	0.646	0.662	0.326
172	-3.922	2.391	0.734	1.337	0.476	0.255	0.645	0.667	0.327
173	-1.961	2.406	1.192	1.517	0.561	0.276	0.730	0.644	0.314
174	-1.961	2.402	1.034	1.392	0.516	0.271	0.710	0.648	0.315
175	-1.961	2.399	0.890	1.423	0.471	0.269	0.710	0.652	0.316
176	0.000	2.415	1.521	1.567	0.555	0.285	0.802	0.657	0.316
177	0.000	2.411	1.295	1.436	0.510	0.286	0.782	0.658	0.315
178	0.000	2.406	1.096	1.501	0.465	0.285	0.784	0.661	0.314
179	-4.762	2.394	1.492	1.612	0.559	0.288	0.782	0.639	0.323
180	-4.762	2.390	1.276	1.479	0.514	0.285	0.761	0.640	0.323
181	-4.762	2.387	1.085	1.521	0.469	0.287	0.761	0.644	0.323
182	3.030	2.484	0.988	1.410	0.563	0.266	0.684	0.685	0.335
183	3.030	2.479	0.863	1.293	0.517	0.261	0.665	0.687	0.335
184	3.030	2.475	0.747	1.319	0.471	0.256	0.665	0.690	0.335
185	-1.961	2.460	0.971	1.448	0.566	0.267	0.665	0.656	0.319
186	-1.961	2.456	0.851	1.329	0.520	0.263	0.646	0.650	0.316
187	-1.961	2.452	0.739	1.339	0.474	0.257	0.645	0.660	0.319
188	0.000	2.469	1.201	1.508	0.559	0.277	0.729	0.634	0.305
189	0.000	2.465	1.040	1.384	0.514	0.274	0.709	0.636	0.305
190	0.000	2.460	0.894	1.423	0.469	0.271	0.710	0.648	0.309
191	1.961	2.477	1.533	1.557	0.554	0.282	0.801	0.655	0.322

192	1.961	2.473	1.302	1.425	0.508	0.288	0.781	0.655	0.321
193	-4.762	2.447	1.181	1.549	0.563	0.278	0.710	0.609	0.310
194	-4.762	2.444	1.026	1.423	0.518	0.274	0.690	0.607	0.308
195	-4.762	2.440	0.886	1.443	0.473	0.272	0.689	0.610	0.309
196	-2.857	2.455	1.502	1.600	0.558	0.285	0.780	0.629	0.314
197	-2.857	2.451	1.283	1.467	0.513	0.288	0.759	0.637	0.316
198	3.030	2.530	0.627	1.230	0.474	0.243	0.608	0.698	0.339
199	-1.961	2.507	0.620	1.248	0.477	0.245	0.589	0.673	0.324
200	0.000	2.523	0.980	1.442	0.565	0.270	0.664	0.645	0.309
201	0.000	2.519	0.857	1.322	0.519	0.266	0.645	0.641	0.307
202	0.000	2.515	0.744	1.339	0.473	0.260	0.645	0.655	0.312
203	1.961	2.532	1.211	1.500	0.558	0.279	0.728	0.627	0.310
204	1.961	2.528	1.047	1.375	0.513	0.276	0.709	0.628	0.309
205	1.961	2.524	0.899	1.422	0.468	0.273	0.710	0.635	0.311
206	3.922	2.541	1.544	1.546	0.552	0.282	0.799	0.652	0.328
207	-4.762	2.502	0.964	1.480	0.568	0.270	0.647	0.617	0.313
208	-4.762	2.498	0.846	1.359	0.522	0.267	0.628	0.616	0.311
209	-4.762	2.494	0.736	1.359	0.476	0.262	0.626	0.621	0.313
210	-2.857	2.510	1.190	1.540	0.562	0.279	0.709	0.605	0.303
211	-2.857	2.506	1.032	1.414	0.517	0.274	0.689	0.603	0.301
212	-2.857	2.502	0.890	1.442	0.471	0.274	0.689	0.606	0.302
213	-0.952	2.518	1.512	1.589	0.556	0.286	0.778	0.620	0.305
214	-0.952	2.514	1.288	1.456	0.511	0.287	0.757	0.628	0.307
215	0.000	2.570	0.625	1.251	0.476	0.249	0.589	0.670	0.318
216	1.961	2.588	0.989	1.436	0.563	0.271	0.664	0.637	0.313
217	1.961	2.583	0.864	1.317	0.517	0.268	0.645	0.634	0.311
218	1.961	2.579	0.749	1.340	0.471	0.264	0.645	0.646	0.315
219	3.922	2.597	1.221	1.492	0.556	0.283	0.727	0.622	0.315
220	3.922	2.592	1.054	1.367	0.511	0.279	0.708	0.622	0.314
221	3.922	2.588	0.903	1.422	0.466	0.273	0.710	0.628	0.315
222	-4.762	2.549	0.619	1.270	0.479	0.250	0.572	0.655	0.327
223	-2.857	2.565	0.972	1.473	0.566	0.272	0.646	0.613	0.306
224	-2.857	2.561	0.852	1.352	0.521	0.269	0.627	0.611	0.304
225	-2.857	2.557	0.741	1.360	0.475	0.265	0.626	0.617	0.306
226	-0.952	2.573	1.198	1.532	0.560	0.283	0.708	0.601	0.296
227	-0.952	2.569	1.038	1.405	0.515	0.279	0.688	0.599	0.294
228	-0.952	2.565	0.893	1.441	0.470	0.275	0.689	0.603	0.295

229	0.952	2.582	1.523	1.578	0.555	0.286	0.776	0.614	0.303
230	-3.704	2.560	1.494	1.622	0.559	0.289	0.757	0.590	0.302
231	1.961	2.635	0.630	1.253	0.475	0.254	0.589	0.667	0.324
232	3.922	2.653	0.997	1.430	0.562	0.273	0.663	0.631	0.318
233	3.922	2.649	0.870	1.311	0.516	0.270	0.645	0.629	0.316
234	3.922	2.644	0.753	1.341	0.470	0.266	0.646	0.637	0.319
235	-2.857	2.612	0.623	1.272	0.478	0.255	0.572	0.653	0.321
236	-0.952	2.629	0.980	1.467	0.565	0.274	0.646	0.609	0.299
237	-0.952	2.625	0.858	1.346	0.519	0.271	0.627	0.608	0.297
238	-0.952	2.621	0.745	1.360	0.473	0.267	0.626	0.613	0.299
239	0.952	2.638	1.208	1.523	0.559	0.286	0.707	0.598	0.296
240	0.952	2.633	1.045	1.397	0.514	0.281	0.688	0.596	0.294
241	0.952	2.629	0.897	1.441	0.468	0.278	0.689	0.599	0.294
242	-3.704	2.615	1.187	1.564	0.562	0.286	0.689	0.583	0.298
243	-3.704	2.611	1.031	1.436	0.517	0.280	0.670	0.581	0.296
244	-3.704	2.607	0.889	1.460	0.472	0.280	0.669	0.584	0.297
245	3.922	2.706	0.729	1.246	0.521	0.263	0.591	0.660	0.329
246	3.922	2.701	0.635	1.256	0.474	0.254	0.590	0.664	0.330
247	-0.952	2.681	0.719	1.278	0.524	0.265	0.574	0.647	0.314
248	-0.952	2.677	0.628	1.274	0.477	0.255	0.572	0.650	0.314
249	0.952	2.694	0.989	1.461	0.563	0.276	0.645	0.605	0.298
250	0.952	2.690	0.864	1.340	0.518	0.273	0.627	0.604	0.297
251	0.952	2.686	0.750	1.361	0.472	0.269	0.627	0.609	0.298
252	2.857	2.704	1.217	1.515	0.557	0.286	0.706	0.595	0.302
253	2.857	2.699	1.051	1.389	0.512	0.284	0.687	0.593	0.300
254	2.857	2.694	0.901	1.439	0.467	0.279	0.688	0.596	0.300
255	-3.704	2.671	0.973	1.498	0.567	0.276	0.629	0.590	0.301
256	-3.704	2.667	0.853	1.375	0.521	0.274	0.610	0.589	0.299
257	-3.704	2.663	0.742	1.380	0.475	0.271	0.609	0.593	0.300
258	-1.852	2.680	1.195	1.554	0.561	0.286	0.688	0.580	0.292
259	-1.852	2.676	1.036	1.426	0.516	0.281	0.669	0.578	0.290
260	-1.852	2.672	0.892	1.459	0.471	0.282	0.668	0.581	0.290
261	0.952	2.747	0.725	1.274	0.522	0.268	0.574	0.645	0.314
262	0.952	2.743	0.633	1.276	0.475	0.261	0.572	0.648	0.315
263	2.857	2.761	0.997	1.454	0.562	0.278	0.645	0.602	0.304
264	2.857	2.756	0.870	1.334	0.516	0.275	0.627	0.600	0.303
265	2.857	2.752	0.754	1.362	0.471	0.270	0.627	0.605	0.304

266	4.762	2.771	1.226	1.507	0.556	0.288	0.705	0.592	0.308
267	4.762	2.766	1.057	1.381	0.511	0.284	0.687	0.590	0.306
268	4.762	2.761	0.905	1.439	0.465	0.281	0.688	0.594	0.307
269	-3.704	2.724	0.716	1.307	0.525	0.269	0.558	0.613	0.309
270	-3.704	2.720	0.626	1.295	0.478	0.262	0.556	0.621	0.312
271	-1.852	2.737	0.981	1.491	0.565	0.278	0.628	0.587	0.294
272	-1.852	2.732	0.859	1.369	0.520	0.276	0.610	0.586	0.293
273	-1.852	2.728	0.746	1.381	0.474	0.272	0.609	0.590	0.294
274	0.000	2.746	1.204	1.546	0.560	0.289	0.687	0.578	0.286
275	0.000	2.741	1.042	1.418	0.514	0.284	0.668	0.576	0.284
276	0.000	2.737	0.895	1.458	0.469	0.281	0.668	0.578	0.284
277	-4.505	2.723	1.184	1.586	0.563	0.290	0.670	0.568	0.296
278	-4.505	2.719	1.029	1.456	0.518	0.283	0.651	0.565	0.294
279	-4.505	2.715	0.887	1.477	0.473	0.284	0.650	0.566	0.294
280	2.857	2.814	0.731	1.270	0.521	0.269	0.574	0.642	0.320
281	2.857	2.810	0.637	1.278	0.474	0.261	0.573	0.645	0.321
282	4.762	2.829	1.005	1.448	0.561	0.279	0.644	0.599	0.310
283	4.762	2.824	0.876	1.327	0.515	0.276	0.627	0.597	0.309
284	4.762	2.819	0.758	1.362	0.469	0.271	0.627	0.602	0.310
285	-1.852	2.790	0.722	1.302	0.524	0.270	0.558	0.609	0.303
286	-1.852	2.786	0.631	1.296	0.477	0.262	0.556	0.616	0.305
287	0.000	2.803	0.988	1.484	0.564	0.279	0.627	0.585	0.288
288	0.000	2.799	0.864	1.362	0.518	0.277	0.609	0.583	0.287
289	0.000	2.794	0.750	1.381	0.473	0.273	0.609	0.587	0.288
290	1.852	2.813	1.212	1.537	0.558	0.289	0.686	0.575	0.292
291	1.852	2.808	1.048	1.409	0.513	0.286	0.667	0.573	0.290
292	1.852	2.803	0.899	1.456	0.468	0.283	0.668	0.576	0.290
293	-4.505	2.780	0.973	1.521	0.567	0.280	0.612	0.575	0.299
294	-4.505	2.776	0.854	1.397	0.522	0.278	0.593	0.573	0.297
295	-4.505	2.772	0.743	1.400	0.476	0.274	0.592	0.574	0.297
296	-2.703	2.789	1.192	1.576	0.562	0.290	0.669	0.565	0.290
297	-2.703	2.785	1.034	1.447	0.517	0.286	0.650	0.563	0.288
298	-2.703	2.781	0.890	1.475	0.471	0.286	0.649	0.564	0.288
299	4.762	2.883	0.737	1.267	0.520	0.273	0.574	0.639	0.326
300	4.762	2.878	0.642	1.280	0.473	0.266	0.573	0.642	0.327
301	0.000	2.857	0.728	1.298	0.523	0.274	0.558	0.605	0.296
302	0.000	2.852	0.635	1.298	0.476	0.267	0.556	0.611	0.298

303	1.852	2.871	0.996	1.478	0.563	0.281	0.627	0.582	0.294
304	1.852	2.866	0.870	1.355	0.517	0.278	0.609	0.580	0.293
305	1.852	2.862	0.754	1.381	0.471	0.275	0.609	0.584	0.294
306	3.704	2.881	1.222	1.529	0.557	0.291	0.685	0.573	0.298
307	3.704	2.876	1.054	1.401	0.512	0.288	0.667	0.571	0.296
308	3.704	2.871	0.903	1.456	0.466	0.286	0.668	0.573	0.296
309	-4.505	2.834	0.719	1.331	0.525	0.275	0.543	0.589	0.304
310	-4.505	2.830	0.629	1.317	0.479	0.268	0.541	0.593	0.305
311	-2.703	2.847	0.980	1.514	0.566	0.281	0.611	0.573	0.293
312	-2.703	2.842	0.859	1.390	0.520	0.278	0.593	0.570	0.291
313	-2.703	2.838	0.747	1.400	0.475	0.276	0.592	0.572	0.291
314	-0.901	2.856	1.201	1.568	0.561	0.290	0.668	0.563	0.284
315	-0.901	2.851	1.040	1.438	0.515	0.288	0.649	0.561	0.282
316	-0.901	2.847	0.894	1.474	0.470	0.286	0.649	0.561	0.282
317	1.852	2.930	0.833	1.411	0.568	0.280	0.575	0.598	0.301
318	1.852	2.925	0.733	1.293	0.521	0.275	0.558	0.602	0.302
319	1.852	2.921	0.639	1.299	0.475	0.267	0.557	0.607	0.303
320	3.704	2.940	1.004	1.472	0.561	0.281	0.627	0.579	0.300
321	3.704	2.935	0.876	1.349	0.516	0.279	0.609	0.577	0.299
322	3.704	2.930	0.758	1.381	0.470	0.276	0.609	0.581	0.300
323	-2.703	2.905	0.821	1.445	0.571	0.281	0.560	0.587	0.299
324	-2.703	2.901	0.724	1.325	0.524	0.276	0.543	0.587	0.298
325	-2.703	2.897	0.633	1.318	0.478	0.269	0.541	0.591	0.299
326	-0.901	2.914	0.988	1.507	0.564	0.282	0.611	0.570	0.287
327	-0.901	2.910	0.865	1.384	0.519	0.281	0.593	0.568	0.285
328	-0.901	2.906	0.751	1.400	0.473	0.277	0.592	0.569	0.285
329	0.901	2.924	1.208	1.559	0.559	0.291	0.667	0.561	0.285
330	0.901	2.919	1.045	1.429	0.514	0.287	0.648	0.559	0.283
331	0.901	2.914	0.896	1.473	0.468	0.288	0.649	0.559	0.282
332	-3.509	2.900	1.189	1.598	0.563	0.292	0.651	0.551	0.288
333	-3.509	2.896	1.031	1.466	0.517	0.287	0.632	0.548	0.287
334	-3.509	2.892	0.888	1.492	0.472	0.290	0.631	0.549	0.286

АНАЛИЗ РЕЗУЛЬТАТОВ

(с указанием номеров вариантов в нисходящей последовательности)

Лучшие варианты относительно передаточного отношения: 15 16 25 26 27 31 32 33 40 95 ...
 Лучшие варианты относительно наименьшего веса: 11 10 9 14 13 2 12 16 5 15 ...
 Оптимальные варианты относительно трения (AC/AE): 46 121 8 78 157 36 109 65 142 27 ...
 Лучшие варианты относительно вариации жесткости: 1 2 17 21 29 30 40 54 59 70 ...
 Лучшие варианты относительно прочности: 333 334 332 330 331 315 329 316 314 297 ...
 Лучшие варианты в общем (сумма) : 316 331 315 330 275 276 314 329 328 327 ...

ОПРЕДЕЛЕНИЕ ВАРИАНТОВ ЗУБЧАТОГО ЗАЦЕПЛЕНИЯ

Вводные данные!

Заданное передаточное отношение 1.500 +/- 5.000 % (1.425 ... 1.575)

Мощность: 32.7145 kW

Исходная частота вращения: 1562.00 (Колесо 1)

Модуль 2.0000 mm

Угол зацепления 20.000

Угол наклона линии зуба 23.000

Межосевое расстояние от 80.0000 mm до 100.0000 mm с шагом 1.0000 mm

Смещение исходного контура допустимо в диапазоне -0.500 ... 0.500

Минимальное число зубьев 18

>> Отбрасываются результаты с удельным скольжением выше (zeta > 3.000; zeta < -3.000).

>> Результаты толщины зуба на головке < минимальная толщина зуба отбрасываются.

x1(max)-criteria: согласно скомпенсированному удельному скольжению

>> Момент инерции: Выдаются данные для области делительной окружности до внутреннего диаметра (при внутреннем зацеплении для делительной окружности до наружного диаметра)

>> Варианты, не отвечающие заданным запасам прочности, отклоняются.

Комментарий:

№ = номер варианта

a = Межосевое расстояние

m = нормальный модуль

alfa = Угол зацепления

бета = угол наклона линии зуба

Z1, Z2 = число зубьев колес 1, 2

X1+X2, X1, X2 = смещение исходного контура, сумма, колесо 1, колесо 2
 Ea = Торцевое перекрытие $(\epsilon_{ps_a.e} + \epsilon_{ps_a.i})/2$
 Eb = Осовое перекрытие
 Eg = общее перекрытие $(\epsilon_{ps_g.e} + \epsilon_{ps_g.i})/2$
 * = Варианты, отмеченные *, являются оптимальными (целочисленные общие перекрытия)
 слайд = удельное скольжение (максимальное значение)
 i = передаточное отношение
 diff_i = отклонение от заданного передаточного отношения в %
 SF1, SF2 = запас прочности ножки зуба колес 1, 2
 SF1, SF2 = запас прочности боковой поверхности (в однопарном зацеплении) колес 1, 2
 SB = запас прочности на заедание, температура «вспышки» в пятне контакта
 SInt = запас прочности на заедание, интегральная температура
 del_cg = Стандартное отклонение жесткости при обкатке (Н/мм/мум)

No.	a	mn	alfa	beta	Z1	Z2	X1+X2	X1	Ea	Eb	Eg	Slide	i	diff_i	SF1						
SF2	SH1	SH2SB			SInt	del_cg															
1	80.00	2.00	20.0	23.0	28	44	+0.952	+0.458	1.32	1.43	2.75	-0.72-0.72	1.57+4.8		1.79	1.75	1.06	1.07	4.96	3.28	0.19
2	80.00	2.00	20.0	23.0	29	43	+0.952	+0.460	1.32	1.43	2.75	-0.71-0.71	1.48-1.1		1.83	1.79	1.08	1.09	5.12	3.31	0.20
3	80.00	2.00	20.0	23.0	29	44	+0.357	+0.033	1.44	1.43	2.87	-1.18-0.69	1.52+1.1		1.86	1.90	1.08	1.10	3.91	3.20	0.24
4							+0.133	1.44	1.43	2.87	-1.02-0.79				1.88	1.88	1.08	1.09	4.28	3.21	0.23
5							+0.233	1.43	1.43	2.86	-0.88-0.88				1.90	1.86	1.08	1.09	4.60	3.21	0.23
6	80.00	2.00	20.0	23.0	29	45	-0.192	-0.163	1.54	1.43	2.97	-1.54-0.87	1.55+3.4		1.93	2.01	1.07	1.08	3.51	3.08	0.26
7							-0.063	1.53	1.43	2.96	-1.31-0.98				1.96	1.98	1.07	1.08	3.85	3.08	0.26
8							+0.037	1.53	1.43	2.96	-1.11-1.11				1.99	1.95	1.07	1.08	4.09	3.08	0.25
9	80.00	2.00	20.0	23.0	30	43	+0.357	+0.025	1.44	1.43	2.87	-1.16-0.68	1.43-4.4		1.91	1.95	1.10	1.11	4.01	3.23	0.24
10							+0.125	1.44	1.43	2.87	-1.01-0.77				1.93	1.93	1.10	1.11	4.39	3.24	0.23
11							+0.225	1.43	1.43	2.86	-0.87-0.87				1.95	1.91	1.10	1.11	4.74	3.24	0.23
12	80.00	2.00	20.0	23.0	30	44	-0.192	-0.180	1.54	1.43	2.97	-1.50-0.85	1.47-2.2		1.97	2.06	1.09	1.10	3.61	3.11	0.26
13							-0.080	1.54	1.43	2.97	-1.29-0.97				2.01	2.03	1.09	1.10	3.95	3.11	0.26
14							+0.020	1.53	1.43	2.96	-1.09-1.09				2.04	2.00	1.09	1.10	4.22	3.11	0.26
15	80.00	2.00	20.0	23.0	30	45	-0.693	-0.353	1.63	1.43	3.06*	-2.07-1.09	1.50+0.0		2.06	2.19	1.06	1.08	3.16	2.94	0.28
16							-0.253	1.62	1.43	3.05*	-1.71-1.25				2.11	2.14	1.06	1.07	3.48	2.94	0.28
17	81.00	2.00	20.0	23.0	29	44	+0.902	+0.440	1.33	1.43	2.76	-0.72-0.72	1.52+1.1		1.84	1.80	1.08	1.10	5.12	3.32	0.20
18	81.00	2.00	20.0	23.0	29	45	+0.312	+0.019	1.45	1.43	2.88	-1.20-0.70	1.55+3.4		1.87	1.91	1.09	1.10	3.92	3.21	0.24

[illegible]

53	84.00	2.00	20.0	23.0	30	46	+0.756	+0.282	1.37	1.43	2.80	-0.84	-0.65	1.53	+2.2	1.89	1.87	1.12	1.14	4.90	3.38	0.23
54								+0.382	1.37	1.43	2.80	-0.73	-0.73			1.90	1.86	1.12	1.13	5.28	3.38	0.22
55	84.00	2.00	20.0	23.0	30	47	+0.177	-0.032	1.48	1.43	2.91	-1.21	-0.71	1.57	+4.4	1.94	1.98	1.12	1.13	4.07	3.27	0.25
56								+0.068	1.48	1.43	2.91	-1.04	-0.80			1.96	1.96	1.12	1.13	4.46	3.28	0.25
57								+0.168	1.47	1.43	2.90	-0.89	-0.89			1.98	1.94	1.11	1.13	4.76	3.28	0.25
58	84.00	2.00	20.0	23.0	31	45	+0.756	+0.281	1.37	1.43	2.80	-0.83	-0.64	1.45	-3.2	1.93	1.92	1.14	1.15	5.02	3.41	0.23
59								+0.381	1.37	1.43	2.80	-0.72	-0.72			1.94	1.90	1.14	1.15	5.44	3.41	0.22
60	84.00	2.00	20.0	23.0	31	46	+0.177	-0.042	1.48	1.43	2.91	-1.18	-0.70	1.48	-1.1	1.98	2.03	1.14	1.15	4.18	3.30	0.25
61								+0.058	1.48	1.43	2.91	-1.03	-0.79			2.01	2.01	1.14	1.15	4.57	3.31	0.25
62								+0.158	1.47	1.43	2.90	-0.88	-0.88			2.03	1.99	1.14	1.15	4.90	3.31	0.25
63	84.00	2.00	20.0	23.0	31	47	-0.358	-0.235	1.57	1.43	3.00*	-1.53	-0.86	1.52	+1.1	2.06	2.15	1.12	1.14	3.75	3.18	0.28
64								-0.135	1.57	1.43	3.00	-1.30	-0.98			2.10	2.12	1.12	1.14	4.11	3.18	0.27
65								-0.035	1.56	1.43	2.99	-1.10	-1.10			2.13	2.08	1.12	1.13	4.37	3.18	0.27
66	84.00	2.00	20.0	23.0	32	46	-0.358	-0.253	1.57	1.43	3.00*	-1.50	-0.85	1.44	-4.2	2.10	2.21	1.14	1.16	3.84	3.20	0.28
67								-0.153	1.57	1.43	3.00*	-1.28	-0.97			2.14	2.17	1.14	1.15	4.21	3.21	0.27
68								-0.053	1.56	1.43	3.00	-1.09	-1.09			2.18	2.13	1.14	1.15	4.49	3.20	0.27
69	85.00	2.00	20.0	23.0	30	47	+0.708	+0.263	1.38	1.43	2.81	-0.85	-0.66	1.57	+4.4	1.89	1.88	1.13	1.14	4.91	3.39	0.23
70								+0.363	1.38	1.43	2.81	-0.74	-0.73			1.91	1.86	1.12	1.14	5.28	3.39	0.22
71	85.00	2.00	20.0	23.0	31	46	+0.708	+0.262	1.39	1.43	2.82	-0.84	-0.65	1.48	-1.1	1.94	1.92	1.15	1.16	5.03	3.42	0.23
72								+0.362	1.38	1.43	2.81	-0.73	-0.73			1.95	1.91	1.14	1.16	5.44	3.42	0.23
73	85.00	2.00	20.0	23.0	31	47	+0.133	-0.055	1.49	1.43	2.92	-1.20	-0.70	1.52	+1.1	1.99	2.04	1.14	1.16	4.19	3.31	0.26
74								+0.045	1.49	1.43	2.92	-1.03	-0.79			2.01	2.02	1.14	1.16	4.59	3.32	0.25
75								+0.145	1.48	1.43	2.91	-0.89	-0.89			2.03	2.00	1.14	1.15	4.90	3.32	0.25
76	85.00	2.00	20.0	23.0	31	48	-0.398	-0.243	1.58	1.43	3.01*	-1.54	-0.87	1.55	+3.2	2.07	2.17	1.13	1.14	3.77	3.19	0.28
77								-0.143	1.58	1.43	3.01*	-1.31	-0.98			2.11	2.13	1.12	1.14	4.13	3.19	0.27
78								-0.043	1.57	1.43	3.00	-1.11	-1.11			2.14	2.09	1.12	1.14	4.37	3.19	0.27
79	85.00	2.00	20.0	23.0	32	46	+0.133	-0.066	1.49	1.43	2.92	-1.17	-0.69	1.44	-4.2	2.03	2.08	1.16	1.18	4.29	3.34	0.26
80								+0.034	1.49	1.43	2.92	-1.02	-0.78			2.06	2.07	1.16	1.17	4.69	3.35	0.26
81								+0.134	1.48	1.43	2.91	-0.88	-0.88			2.08	2.04	1.16	1.17	5.04	3.34	0.25
82	85.00	2.00	20.0	23.0	32	47	-0.398	-0.262	1.58	1.43	3.01*	-1.51	-0.85	1.47	-2.1	2.11	2.22	1.15	1.16	3.86	3.21	0.28
83								-0.162	1.58	1.43	3.01*	-1.29	-0.97			2.15	2.18	1.14	1.16	4.22	3.22	0.28
84								-0.062	1.57	1.43	3.00*	-1.09	-1.09			2.19	2.14	1.14	1.16	4.50	3.21	0.27
85	86.00	2.00	20.0	23.0	31	47	+0.661	+0.243	1.40	1.43	2.83	-0.84	-0.65	1.52	+1.1	1.94	1.93	1.15	1.17	5.05	3.43	0.24
86								+0.343	1.39	1.43	2.82	-0.73	-0.73			1.96	1.91	1.15	1.16	5.44	3.43	0.23
87	86.00	2.00	20.0	23.0	31	48	+0.089	-0.068	1.50	1.43	2.93	-1.21	-0.71	1.55	+3.2	2.00	2.05	1.15	1.16	4.21	3.32	0.26

88									+0.032	1.50	1.43	2.93	-1.04	-0.80			2.02	2.03	1.15	1.16	4.60	3.33	0.26
89									+0.132	1.49	1.43	2.92	-0.90	-0.90			2.04	2.00	1.14	1.16	4.91	3.33	0.25
90	86.00	2.00	20.0	23.0	32	46	+0.661	+0.241	1.40	1.43	2.83	-0.83	-0.64	1.44-4.2			1.99	1.97	1.17	1.18	5.17	3.45	0.24
91									+0.341	1.39	1.43	2.82	-0.72	-0.72			2.00	1.96	1.17	1.18	5.59	3.45	0.23
92	86.00	2.00	20.0	23.0	32	47	+0.089	-0.079	1.50	1.43	2.93	-1.18	-0.70	1.47-2.1			2.04	2.09	1.17	1.18	4.31	3.35	0.26
93									+0.021	1.50	1.43	2.93	-1.03	-0.79			2.07	2.07	1.17	1.18	4.71	3.36	0.26
94									+0.121	1.49	1.43	2.92	-0.88	-0.88			2.09	2.05	1.16	1.18	5.05	3.35	0.25
95	86.00	2.00	20.0	23.0	32	48	-0.439	-0.270	1.59	1.43	3.02*	-1.52	-0.86	1.50+0.0			2.12	2.23	1.15	1.16	3.87	3.22	0.28
96									-0.170	1.59	1.43	3.02*	-1.30	-0.98			2.16	2.19	1.15	1.16	4.24	3.23	0.28
97									-0.070	1.58	1.43	3.01*	-1.10	-1.10			2.20	2.15	1.14	1.16	4.50	3.22	0.27
98	87.00	2.00	20.0	23.0	31	48	+0.614	+0.126	1.41	1.43	2.84	-0.98	-0.58	1.55+3.2			1.94	1.95	1.16	1.17	4.63	3.43	0.25
99									+0.226	1.41	1.43	2.84	-0.85	-0.66			1.95	1.94	1.16	1.17	5.07	3.44	0.24
100									+0.326	1.40	1.43	2.83	-0.74	-0.74			1.96	1.92	1.15	1.17	5.44	3.44	0.23
101	87.00	2.00	20.0	23.0	32	47	+0.614	+0.123	1.41	1.43	2.84	-0.96	-0.57	1.47-2.1			1.98	2.00	1.18	1.19	4.74	3.45	0.25
102									+0.223	1.41	1.43	2.84	-0.84	-0.65			1.99	1.98	1.18	1.19	5.19	3.46	0.24
103									+0.323	1.40	1.43	2.83	-0.73	-0.73			2.01	1.97	1.17	1.19	5.60	3.46	0.24
104	87.00	2.00	20.0	23.0	32	48	+0.046	-0.092	1.51	1.43	2.94	-1.19	-0.70	1.50+0.0			2.05	2.10	1.17	1.19	4.32	3.36	0.26
105									+0.008	1.51	1.43	2.94	-1.04	-0.79			2.08	2.08	1.17	1.18	4.73	3.36	0.26
106									+0.108	1.50	1.43	2.93	-0.89	-0.89			2.10	2.06	1.17	1.18	5.05	3.36	0.26
107	87.00	2.00	20.0	23.0	32	49	-0.479	-0.278	1.60	1.43	3.03*	-1.54	-0.86	1.53+2.1			2.13	2.24	1.15	1.17	3.89	3.23	0.28
108									-0.178	1.59	1.43	3.02*	-1.31	-0.98			2.18	2.20	1.15	1.17	4.26	3.23	0.28
109									-0.078	1.58	1.43	3.01*	-1.10	-1.10			2.21	2.16	1.15	1.16	4.51	3.23	0.28
110	87.00	2.00	20.0	23.0	33	48	-0.479	-0.298	1.60	1.43	3.03*	-1.50	-0.85	1.45-3.0			2.18	2.29	1.17	1.19	3.98	3.26	0.28
111									-0.198	1.59	1.43	3.02*	-1.28	-0.97			2.22	2.25	1.17	1.18	4.35	3.26	0.28
112									-0.098	1.59	1.43	3.02*	-1.09	-1.09			2.26	2.21	1.17	1.18	4.64	3.26	0.28
113	88.00	2.00	20.0	23.0	32	48	+0.567	+0.105	1.42	1.43	2.85	-0.97	-0.58	1.50+0.0			1.98	2.00	1.18	1.20	4.76	3.46	0.25
114									+0.205	1.42	1.43	2.85	-0.85	-0.66			2.00	1.99	1.18	1.19	5.20	3.47	0.25
115									+0.305	1.41	1.43	2.84	-0.73	-0.73			2.01	1.97	1.18	1.19	5.60	3.47	0.24
116	88.00	2.00	20.0	23.0	32	49	+0.002	-0.105	1.52	1.43	2.95	-1.21	-0.71	1.53+2.1			2.06	2.11	1.18	1.19	4.34	3.37	0.26
117									-0.005	1.51	1.43	2.94	-1.04	-0.80			2.09	2.09	1.17	1.19	4.75	3.37	0.26
118									+0.095	1.51	1.43	2.94	-0.90	-0.90			2.11	2.07	1.17	1.19	5.06	3.37	0.26
119	88.00	2.00	20.0	23.0	32	50	-0.519	-0.286	1.60	1.43	3.03*	-1.55	-0.87	1.56+4.2			2.14	2.25	1.16	1.17	3.90	3.24	0.28
120									-0.186	1.60	1.43	3.03*	-1.31	-0.99			2.19	2.21	1.15	1.17	4.27	3.24	0.28
121									-0.086	1.59	1.43	3.02*	-1.11	-1.11			2.22	2.17	1.15	1.17	4.52	3.24	0.28
122	88.00	2.00	20.0	23.0	33	48	+0.002	-0.116	1.52	1.43	2.95	-1.18	-0.70	1.45-3.0			2.10	2.16	1.19	1.21	4.44	3.39	0.27

123								-0.016	1.52	1.43	2.95	-1.03	-0.79		2.13	2.14	1.19	1.21	4.85	3.40	0.26	
124								+0.084	1.51	1.43	2.94	-0.89	-0.89		2.15	2.11	1.19	1.20	5.19	3.40	0.26	
125	88.00	2.00	20.0	23.0	33	49	-0.519	-0.306	1.60	1.43	3.03*	-1.52	-0.86	1.48	-1.0	2.19	2.30	1.18	1.19	3.99	3.27	0.28
126								-0.206	1.60	1.43	3.03*	-1.29	-0.97		2.23	2.26	1.17	1.19	4.37	3.27	0.28	
127								-0.106	1.59	1.43	3.02*	-1.10	-1.09		2.27	2.22	1.17	1.18	4.64	3.27	0.28	
128	89.00	2.00	20.0	23.0	32	49	+0.520	+0.088	1.43	1.43	2.86	-0.98	-0.59	1.53	+2.1	1.99	2.01	1.19	1.20	4.77	3.47	0.26
129								+0.188	1.43	1.43	2.86	-0.85	-0.66		2.01	1.99	1.18	1.20	5.22	3.48	0.25	
130								+0.288	1.42	1.43	2.85	-0.74	-0.74		2.02	1.98	1.18	1.20	5.60	3.48	0.25	
131	89.00	2.00	20.0	23.0	32	50	-0.041	-0.117	1.53	1.43	2.96	-1.22	-0.71	1.56	+4.2	2.06	2.12	1.18	1.20	4.35	3.37	0.27
132								-0.017	1.52	1.43	2.95	-1.05	-0.80		2.09	2.10	1.18	1.19	4.76	3.38	0.27	
133								+0.083	1.52	1.43	2.95	-0.90	-0.90		2.12	2.07	1.17	1.19	5.06	3.38	0.26	
134	89.00	2.00	20.0	23.0	33	48	+0.520	+0.084	1.43	1.43	2.86	-0.96	-0.58	1.45	-3.0	2.03	2.05	1.21	1.22	4.88	3.49	0.26
135								+0.184	1.43	1.43	2.86	-0.84	-0.65		2.05	2.04	1.20	1.22	5.33	3.51	0.25	
136								+0.284	1.43	1.43	2.86	-0.73	-0.73		2.06	2.02	1.20	1.22	5.75	3.51	0.25	
137	89.00	2.00	20.0	23.0	33	49	-0.041	-0.129	1.53	1.43	2.96	-1.19	-0.70	1.48	-1.0	2.11	2.17	1.20	1.21	4.46	3.40	0.27
138								-0.029	1.52	1.43	2.95	-1.03	-0.79		2.14	2.15	1.20	1.21	4.87	3.41	0.27	
139								+0.071	1.52	1.43	2.95	-0.89	-0.89		2.16	2.12	1.19	1.21	5.20	3.41	0.26	
140	89.00	2.00	20.0	23.0	33	50	-0.559	-0.314	1.61	1.43	3.04*	-1.53	-0.86	1.52	+1.0	2.20	2.31	1.18	1.19	4.01	3.27	0.28
141								-0.214	1.61	1.43	3.04*	-1.30	-0.98		2.24	2.27	1.18	1.19	4.39	3.28	0.28	
142								-0.114	1.60	1.43	3.03*	-1.10	-1.10		2.28	2.23	1.17	1.19	4.65	3.27	0.28	
143	89.00	2.00	20.0	23.0	34	49	-0.559	-0.334	1.61	1.43	3.04*	-1.50	-0.85	1.44	-3.9	2.24	2.36	1.20	1.21	4.10	3.30	0.28
144								-0.234	1.61	1.43	3.04*	-1.28	-0.96		2.29	2.32	1.20	1.21	4.48	3.30	0.28	
145								-0.134	1.60	1.43	3.03*	-1.09	-1.09		2.33	2.28	1.19	1.21	4.77	3.30	0.28	
146	90.00	2.00	20.0	23.0	32	50	+0.474	+0.072	1.44	1.43	2.87	-0.99	-0.59	1.56	+4.2	2.00	2.02	1.19	1.21	4.78	3.48	0.26
147								+0.172	1.44	1.43	2.87	-0.86	-0.67		2.02	2.00	1.19	1.20	5.23	3.49	0.25	
148								+0.272	1.43	1.43	2.86	-0.75	-0.75		2.03	1.99	1.19	1.20	5.61	3.49	0.25	
149	90.00	2.00	20.0	23.0	33	49	+0.474	+0.067	1.44	1.43	2.87	-0.97	-0.58	1.48	-1.0	2.04	2.06	1.21	1.23	4.90	3.50	0.26
150								+0.167	1.44	1.43	2.87	-0.85	-0.66		2.06	2.05	1.21	1.22	5.35	3.52	0.26	
151								+0.267	1.43	1.43	2.86	-0.74	-0.74		2.07	2.03	1.21	1.22	5.76	3.51	0.25	
152	90.00	2.00	20.0	23.0	33	50	-0.083	-0.141	1.54	1.43	2.97	-1.20	-0.71	1.52	+1.0	2.12	2.18	1.20	1.22	4.47	3.41	0.27
153								-0.041	1.53	1.43	2.96	-1.04	-0.80		2.15	2.16	1.20	1.22	4.89	3.42	0.27	
154								+0.059	1.53	1.43	2.96	-0.89	-0.89		2.17	2.13	1.20	1.21	5.21	3.41	0.27	
155	90.00	2.00	20.0	23.0	33	51	-0.598	-0.321	1.62	1.43	3.05*	-1.54	-0.86	1.55	+3.0	2.21	2.32	1.18	1.20	4.02	3.28	0.28
156								-0.221	1.61	1.43	3.04*	-1.31	-0.98		2.25	2.28	1.18	1.19	4.40	3.28	0.28	
157								-0.121	1.60	1.43	3.04*	-1.10	-1.10		2.29	2.24	1.18	1.19	4.66	3.28	0.28	

158	90.00	2.00	20.0	23.0	34	49	-0.083	-0.153	1.54	1.43	2.97	-1.18	-0.70	1.44	-3.9	2.16	2.23	1.22	1.24	4.57	3.43	0.27
159								-0.053	1.53	1.43	2.96	-1.03	-0.79			2.19	2.20	1.22	1.23	4.99	3.44	0.27
160								+0.047	1.53	1.43	2.96	-0.89	-0.89			2.21	2.18	1.22	1.23	5.34	3.44	0.27
161	90.00	2.00	20.0	23.0	34	50	-0.598	-0.342	1.62	1.43	3.05*	-1.51	-0.85	1.47	-2.0	2.25	2.38	1.20	1.22	4.11	3.31	0.28
162								-0.242	1.61	1.43	3.04*	-1.29	-0.97			2.30	2.34	1.20	1.21	4.50	3.31	0.28
163								-0.142	1.61	1.43	3.04*	-1.09	-1.09			2.34	2.29	1.20	1.21	4.78	3.31	0.28
164	91.00	2.00	20.0	23.0	33	50	+0.428	+0.051	1.45	1.43	2.88	-0.98	-0.59	1.52	+1.0	2.05	2.07	1.21	1.23	4.91	3.51	0.26
165								+0.151	1.45	1.43	2.88	-0.86	-0.66			2.07	2.05	1.21	1.23	5.37	3.52	0.26
166								+0.251	1.44	1.43	2.87	-0.74	-0.74			2.08	2.04	1.21	1.23	5.76	3.52	0.25
167	91.00	2.00	20.0	23.0	33	51	-0.126	-0.152	1.54	1.43	2.97	-1.21	-0.71	1.55	+3.0	2.13	2.19	1.21	1.22	4.49	3.42	0.28
168								-0.052	1.54	1.43	2.97	-1.05	-0.80			2.16	2.17	1.20	1.22	4.90	3.42	0.27
169								+0.048	1.53	1.43	2.96	-0.90	-0.90			2.18	2.14	1.20	1.22	5.21	3.42	0.27
170	91.00	2.00	20.0	23.0	34	49	+0.428	+0.046	1.45	1.43	2.88	-0.96	-0.58	1.44	-3.9	2.09	2.11	1.23	1.25	5.02	3.54	0.26
171								+0.146	1.45	1.43	2.88	-0.84	-0.66			2.11	2.10	1.23	1.25	5.48	3.55	0.26
172								+0.246	1.45	1.43	2.88	-0.73	-0.73			2.12	2.08	1.23	1.24	5.91	3.55	0.26
173	91.00	2.00	20.0	23.0	34	50	-0.126	-0.165	1.54	1.43	2.97	-1.19	-0.70	1.47	-2.0	2.17	2.24	1.23	1.24	4.59	3.44	0.28
174								-0.065	1.54	1.43	2.97	-1.03	-0.79			2.20	2.21	1.22	1.24	5.01	3.45	0.27
175								+0.035	1.54	1.43	2.97	-0.89	-0.89			2.22	2.19	1.22	1.23	5.35	3.45	0.27
176	91.00	2.00	20.0	23.0	34	51	-0.638	-0.349	1.63	1.43	3.06*	-1.52	-0.86	1.50	+0.0	2.26	2.39	1.20	1.22	4.12	3.32	0.28
177								-0.249	1.62	1.43	3.05*	-1.29	-0.97			2.31	2.35	1.20	1.22	4.52	3.32	0.29
178								-0.149	1.61	1.43	3.04*	-1.10	-1.10			2.35	2.30	1.20	1.21	4.79	3.32	0.28
179	91.00	2.00	20.0	23.0	35	50	-0.638	-0.370	1.63	1.43	3.06*	-1.49	-0.84	1.43	-4.8	2.31	2.44	1.22	1.24	4.21	3.34	0.29
180								-0.270	1.62	1.43	3.05*	-1.28	-0.96			2.36	2.40	1.22	1.23	4.60	3.34	0.28
181								-0.170	1.62	1.43	3.05*	-1.09	-1.08			2.40	2.35	1.22	1.23	4.91	3.34	0.29
182	92.00	2.00	20.0	23.0	33	51	+0.382	+0.035	1.46	1.43	2.89	-0.99	-0.59	1.55	+3.0	2.05	2.08	1.22	1.24	4.93	3.52	0.27
183								+0.135	1.46	1.43	2.89	-0.86	-0.67			2.07	2.06	1.22	1.23	5.39	3.53	0.26
184								+0.235	1.45	1.43	2.88	-0.75	-0.75			2.09	2.04	1.21	1.23	5.76	3.53	0.26
185	92.00	2.00	20.0	23.0	34	50	+0.382	+0.030	1.46	1.43	2.89	-0.97	-0.58	1.47	-2.0	2.10	2.12	1.24	1.25	5.04	3.54	0.27
186								+0.130	1.46	1.43	2.89	-0.85	-0.66			2.11	2.10	1.24	1.25	5.50	3.56	0.26
187								+0.230	1.45	1.43	2.88	-0.74	-0.74			2.13	2.09	1.23	1.25	5.91	3.55	0.26
188	92.00	2.00	20.0	23.0	34	51	-0.168	-0.177	1.55	1.43	2.98	-1.20	-0.70	1.50	+0.0	2.18	2.25	1.23	1.24	4.60	3.45	0.28
189								-0.077	1.55	1.43	2.98	-1.04	-0.80			2.21	2.22	1.23	1.24	5.03	3.46	0.27
190								+0.023	1.54	1.43	2.97	-0.89	-0.89			2.23	2.19	1.22	1.24	5.36	3.45	0.27
191	92.00	2.00	20.0	23.0	34	52	-0.677	-0.356	1.63	1.43	3.06*	-1.53	-0.86	1.53	+2.0	2.27	2.40	1.21	1.22	4.14	3.32	0.28
192								-0.256	1.63	1.43	3.06*	-1.30	-0.98			2.32	2.36	1.20	1.22	4.53	3.33	0.29

193	92.00	2.00	20.0	23.0	35	50	-0.168	-0.190	1.55	1.43	2.98	-1.18	-0.69	1.43	4.8	2.22	2.29	1.25	1.26	4.70	3.47	0.28
194								-0.090	1.55	1.43	2.98	-1.03	-0.79			2.25	2.27	1.25	1.26	5.13	3.48	0.27
195								+0.010	1.54	1.43	2.98	-0.89	-0.89			2.28	2.24	1.24	1.26	5.49	3.48	0.27
196	92.00	2.00	20.0	23.0	35	51	-0.677	-0.377	1.63	1.43	3.06*	-1.50	-0.85	1.46	-2.9	2.32	2.45	1.23	1.24	4.23	3.35	0.28
197								-0.277	1.63	1.43	3.06*	-1.28	-0.96			2.37	2.41	1.22	1.24	4.62	3.35	0.29
198	93.00	2.00	20.0	23.0	33	51	+0.923	+0.439	1.36	1.43	2.79	-0.63	-0.63	1.55	+3.0	2.01	1.97	1.22	1.23	6.34	3.62	0.24
199	93.00	2.00	20.0	23.0	34	50	+0.923	+0.441	1.36	1.43	2.79	-0.62	-0.62	1.47	-2.0	2.05	2.01	1.24	1.25	6.50	3.64	0.25
200	93.00	2.00	20.0	23.0	34	51	+0.337	+0.014	1.47	1.43	2.90	-0.98	-0.59	1.50	+0.0	2.10	2.13	1.24	1.26	5.06	3.55	0.27
201								+0.114	1.47	1.43	2.90	-0.86	-0.66			2.12	2.11	1.24	1.26	5.52	3.56	0.27
202								+0.214	1.46	1.43	2.89	-0.74	-0.74			2.14	2.09	1.24	1.25	5.92	3.56	0.26
203	93.00	2.00	20.0	23.0	34	52	-0.210	-0.188	1.56	1.43	2.99	-1.21	-0.71	1.53	+2.0	2.19	2.26	1.23	1.25	4.62	3.46	0.28
204								-0.088	1.56	1.43	2.99	-1.05	-0.80			2.22	2.23	1.23	1.25	5.04	3.46	0.28
205								+0.012	1.55	1.43	2.98	-0.90	-0.90			2.24	2.20	1.23	1.24	5.36	3.46	0.27
206	93.00	2.00	20.0	23.0	34	53	-0.716	-0.363	1.64	1.43	3.07*	-1.54	-0.86	1.56	+3.9	2.28	2.41	1.21	1.23	4.15	3.33	0.28
207	93.00	2.00	20.0	23.0	35	50	+0.337	+0.008	1.47	1.43	2.90	-0.96	-0.58	1.43	-4.8	2.14	2.17	1.26	1.27	5.16	3.57	0.27
208								+0.108	1.47	1.43	2.90	-0.85	-0.66			2.16	2.15	1.26	1.27	5.63	3.59	0.27
209								+0.208	1.46	1.43	2.89	-0.74	-0.74			2.18	2.14	1.26	1.27	6.06	3.58	0.26
210	93.00	2.00	20.0	23.0	35	51	-0.210	-0.202	1.56	1.43	2.99	-1.19	-0.70	1.46	-2.9	2.23	2.31	1.25	1.27	4.72	3.48	0.28
211								-0.102	1.56	1.43	2.99	-1.03	-0.79			2.26	2.28	1.25	1.26	5.15	3.49	0.27
212								-0.002	1.55	1.43	2.98	-0.89	-0.89			2.29	2.25	1.25	1.26	5.50	3.48	0.27
213	93.00	2.00	20.0	23.0	35	52	-0.716	-0.384	1.64	1.43	3.07*	-1.51	-0.85	1.49	-1.0	2.33	2.46	1.23	1.24	4.24	3.35	0.29
214								-0.284	1.63	1.43	3.06*	-1.29	-0.97			2.38	2.42	1.23	1.24	4.64	3.36	0.29
215	94.00	2.00	20.0	23.0	34	51	+0.875	+0.421	1.37	1.43	2.81	-0.63	-0.63	1.50	+0.0	2.06	2.01	1.24	1.26	6.50	3.65	0.25
216	94.00	2.00	20.0	23.0	34	52	+0.292	-0.002	1.48	1.43	2.91	-0.99	-0.59	1.53	+2.0	2.11	2.14	1.25	1.26	5.07	3.56	0.27
217								+0.098	1.48	1.43	2.91	-0.86	-0.67			2.13	2.12	1.24	1.26	5.54	3.57	0.27
218								+0.198	1.47	1.43	2.90	-0.75	-0.75			2.15	2.10	1.24	1.26	5.92	3.57	0.26
219	94.00	2.00	20.0	23.0	34	53	-0.252	-0.199	1.57	1.43	3.00	-1.22	-0.71	1.56	+3.9	2.20	2.27	1.24	1.25	4.63	3.47	0.28
220								-0.099	1.56	1.43	2.99	-1.05	-0.81			2.23	2.24	1.23	1.25	5.06	3.47	0.28
221								+0.001	1.56	1.43	2.99	-0.90	-0.90			2.25	2.21	1.23	1.25	5.37	3.47	0.27
222	94.00	2.00	20.0	23.0	35	50	+0.875	+0.423	1.38	1.43	2.81	-0.62	-0.62	1.43	-4.8	2.10	2.05	1.26	1.28	6.65	3.67	0.25
223	94.00	2.00	20.0	23.0	35	51	+0.292	-0.008	1.48	1.43	2.91	-0.97	-0.58	1.46	-2.9	2.15	2.18	1.27	1.28	5.18	3.58	0.27
224								+0.092	1.48	1.43	2.91	-0.85	-0.66			2.17	2.16	1.26	1.28	5.65	3.59	0.27
225								+0.192	1.47	1.43	2.90	-0.74	-0.74			2.19	2.15	1.26	1.28	6.07	3.59	0.27
226	94.00	2.00	20.0	23.0	35	52	-0.252	-0.213	1.57	1.43	3.00	-1.20	-0.70	1.49	-1.0	2.24	2.32	1.26	1.27	4.73	3.49	0.28
227								-0.113	1.56	1.43	3.00	-1.04	-0.80			2.27	2.29	1.25	1.27	5.17	3.50	0.28

228							-0.013	1.56	1.43	2.99	-0.89	-0.89		2.30	2.26	1.25	1.26	5.50	3.49	0.28			
229	94.00	2.00	20.0	23.0	35	53	-0.755	-0.391	1.64	1.43	3.07*	-1.52	-0.85	1.51	+1.0		2.34	2.48	1.23	1.25	4.26	3.36	0.29
230	94.00	2.00	20.0	23.0	36	52	-0.755	-0.412	1.64	1.43	3.07*	-1.49	-0.84	1.44	-3.7		2.39	2.53	1.25	1.26	4.34	3.38	0.29
231	95.00	2.00	20.0	23.0	34	52	+0.827	+0.402	1.38	1.43	2.81	-0.63	-0.63	1.53	+2.0		2.07	2.02	1.25	1.26	6.50	3.66	0.25
232	95.00	2.00	20.0	23.0	34	53	+0.247	-0.017	1.49	1.43	2.92	-1.00	-0.60	1.56	+3.9		2.12	2.15	1.25	1.27	5.09	3.57	0.27
233							+0.083	1.49	1.43	2.92	-0.87	-0.67					2.14	2.13	1.25	1.27	5.55	3.58	0.27
234							+0.183	1.48	1.43	2.91	-0.75	-0.75					2.15	2.11	1.25	1.26	5.93	3.58	0.27
235	95.00	2.00	20.0	23.0	35	51	+0.827	+0.402	1.39	1.43	2.82	-0.62	-0.62	1.46	-2.9		2.10	2.06	1.27	1.28	6.66	3.68	0.26
236	95.00	2.00	20.0	23.0	35	52	+0.247	-0.023	1.49	1.43	2.92	-0.98	-0.59	1.49	-1.0		2.16	2.19	1.27	1.29	5.20	3.59	0.27
237							+0.077	1.49	1.43	2.92	-0.86	-0.67					2.18	2.17	1.27	1.28	5.67	3.60	0.27
238							+0.177	1.48	1.43	2.91	-0.75	-0.75					2.19	2.15	1.27	1.28	6.08	3.60	0.27
239	95.00	2.00	20.0	23.0	35	53	-0.294	-0.224	1.58	1.43	3.01*	-1.21	-0.71	1.51	+1.0		2.25	2.33	1.26	1.27	4.75	3.50	0.29
240							-0.124	1.57	1.43	3.00*	-1.04	-0.80					2.28	2.30	1.26	1.27	5.18	3.50	0.28
241							-0.024	1.57	1.43	3.00	-0.90	-0.90					2.31	2.27	1.25	1.27	5.51	3.50	0.28
242	95.00	2.00	20.0	23.0	36	52	-0.294	-0.238	1.58	1.43	3.01*	-1.19	-0.70	1.44	-3.7		2.29	2.37	1.28	1.29	4.85	3.52	0.29
243							-0.138	1.57	1.43	3.00*	-1.03	-0.79					2.32	2.35	1.27	1.29	5.28	3.52	0.28
244							-0.038	1.57	1.43	3.00	-0.89	-0.89					2.35	2.31	1.27	1.29	5.64	3.52	0.28
245	96.00	2.00	20.0	23.0	34	53	+0.779	+0.282	1.40	1.43	2.83	-0.73	-0.57	1.56	+3.9		2.06	2.04	1.25	1.27	6.05	3.66	0.26
246							+0.382	1.39	1.43	2.82	-0.64	-0.63					2.07	2.02	1.25	1.27	6.50	3.66	0.25
247	96.00	2.00	20.0	23.0	35	52	+0.779	+0.283	1.40	1.43	2.83	-0.72	-0.56	1.49	-1.0		2.10	2.08	1.27	1.29	6.17	3.69	0.27
248							+0.383	1.40	1.43	2.83	-0.63	-0.63					2.11	2.07	1.27	1.29	6.66	3.69	0.26
249	96.00	2.00	20.0	23.0	35	53	+0.203	-0.038	1.50	1.43	2.93	-0.99	-0.59	1.51	+1.0		2.17	2.20	1.27	1.29	5.21	3.60	0.28
250							+0.062	1.49	1.43	2.93	-0.86	-0.67					2.19	2.18	1.27	1.29	5.69	3.61	0.27
251							+0.162	1.49	1.43	2.92	-0.75	-0.75					2.20	2.16	1.27	1.29	6.08	3.61	0.27
252	96.00	2.00	20.0	23.0	35	54	-0.335	-0.234	1.58	1.43	3.01*	-1.22	-0.71	1.54	+2.9		2.26	2.34	1.26	1.28	4.76	3.50	0.29
253							-0.134	1.58	1.43	3.01*	-1.05	-0.80					2.29	2.31	1.26	1.28	5.20	3.51	0.28
254							-0.034	1.57	1.43	3.00*	-0.90	-0.90					2.32	2.28	1.26	1.27	5.52	3.51	0.28
255	96.00	2.00	20.0	23.0	36	52	+0.203	-0.046	1.50	1.43	2.93	-0.97	-0.59	1.44	-3.7		2.21	2.24	1.29	1.31	5.32	3.62	0.28
256							+0.054	1.50	1.43	2.93	-0.85	-0.66					2.23	2.22	1.29	1.31	5.80	3.63	0.27
257							+0.154	1.49	1.43	2.92	-0.74	-0.74					2.24	2.20	1.29	1.30	6.22	3.63	0.27
258	96.00	2.00	20.0	23.0	36	53	-0.335	-0.249	1.58	1.43	3.01*	-1.20	-0.70	1.47	-1.9		2.30	2.39	1.28	1.30	4.86	3.53	0.29
259							-0.149	1.58	1.43	3.01*	-1.04	-0.79					2.33	2.36	1.28	1.29	5.31	3.53	0.28
260							-0.049	1.57	1.43	3.00*	-0.89	-0.89					2.36	2.32	1.28	1.29	5.65	3.53	0.28
261	97.00	2.00	20.0	23.0	35	53	+0.732	+0.264	1.41	1.43	2.84	-0.73	-0.57	1.51	+1.0		2.11	2.09	1.28	1.29	6.19	3.69	0.27

262								+0.364	1.40	1.43	2.83	-0.63	-0.63		2.12	2.07	1.28	1.29	6.66	3.69	0.26
263	97.00	2.00	20.0	23.0	35	54	+0.159	-0.053	1.51	1.43	2.94	-1.00	-0.60	1.54+2.9	2.18	2.21	1.28	1.30	5.23	3.61	0.28
264								+0.047	1.50	1.43	2.93	-0.87	-0.67		2.20	2.19	1.28	1.29	5.70	3.62	0.28
265								+0.147	1.50	1.43	2.93	-0.75	-0.75		2.21	2.17	1.27	1.29	6.09	3.62	0.27
266	97.00	2.00	20.0	23.0	35	55	-0.376	-0.244	1.59	1.43	3.02*	-1.23	-0.72	1.57+4.8	2.27	2.35	1.27	1.28	4.78	3.51	0.29
267								-0.144	1.58	1.43	3.01*	-1.06	-0.81		2.30	2.32	1.26	1.28	5.22	3.52	0.28
268								-0.044	1.58	1.43	3.01*	-0.90	-0.90		2.33	2.29	1.26	1.28	5.52	3.51	0.28
269	97.00	2.00	20.0	23.0	36	52	+0.732	+0.263	1.41	1.43	2.84	-0.72	-0.56	1.44-3.7	2.15	2.12	1.30	1.31	6.31	3.71	0.27
270								+0.363	1.41	1.43	2.84	-0.63	-0.63		2.15	2.11	1.29	1.31	6.82	3.71	0.26
271	97.00	2.00	20.0	23.0	36	53	+0.159	-0.060	1.51	1.43	2.94	-0.98	-0.59	1.47-1.9	2.22	2.25	1.30	1.31	5.34	3.63	0.28
272								+0.040	1.50	1.43	2.93	-0.86	-0.67		2.24	2.23	1.29	1.31	5.82	3.64	0.28
273								+0.140	1.50	1.43	2.93	-0.75	-0.75		2.25	2.21	1.29	1.31	6.23	3.64	0.27
274	97.00	2.00	20.0	23.0	36	54	-0.376	-0.259	1.59	1.43	3.02*	-1.20	-0.71	1.50+0.0	2.31	2.40	1.28	1.30	4.88	3.53	0.29
275								-0.159	1.59	1.43	3.02*	-1.04	-0.80		2.35	2.37	1.28	1.30	5.32	3.54	0.28
276								-0.059	1.58	1.43	3.01*	-0.90	-0.90		2.37	2.33	1.28	1.29	5.66	3.54	0.28
277	97.00	2.00	20.0	23.0	37	53	-0.376	-0.274	1.59	1.43	3.02*	-1.18	-0.70	1.43-4.5	2.33	2.44	1.30	1.32	4.97	3.55	0.29
278								-0.174	1.59	1.43	3.02*	-1.03	-0.79		2.37	2.41	1.30	1.31	5.42	3.56	0.28
279								-0.074	1.58	1.43	3.01*	-0.89	-0.89		2.40	2.38	1.30	1.31	5.79	3.56	0.28
280	98.00	2.00	20.0	23.0	35	54	+0.684	+0.245	1.42	1.43	2.85	-0.73	-0.57	1.54+2.9	2.12	2.09	1.28	1.30	6.21	3.70	0.27
281								+0.345	1.41	1.43	2.84	-0.64	-0.64		2.13	2.08	1.28	1.30	6.67	3.70	0.26
282	98.00	2.00	20.0	23.0	35	55	+0.115	-0.067	1.51	1.43	2.94	-1.00	-0.60	1.57+4.8	2.18	2.22	1.28	1.30	5.24	3.61	0.28
283								+0.033	1.51	1.43	2.94	-0.88	-0.68		2.21	2.20	1.28	1.30	5.72	3.62	0.28
284								+0.133	1.50	1.43	2.93	-0.76	-0.76		2.22	2.18	1.28	1.29	6.09	3.62	0.27
285	98.00	2.00	20.0	23.0	36	53	+0.684	+0.244	1.42	1.43	2.85	-0.72	-0.56	1.47-1.9	2.15	2.13	1.30	1.32	6.33	3.72	0.27
286								+0.344	1.42	1.43	2.85	-0.63	-0.63		2.16	2.12	1.30	1.31	6.82	3.72	0.26
287	98.00	2.00	20.0	23.0	36	54	+0.115	-0.075	1.51	1.43	2.94	-0.99	-0.59	1.50+0.0	2.22	2.26	1.30	1.32	5.35	3.64	0.28
288								+0.025	1.51	1.43	2.94	-0.86	-0.67		2.25	2.24	1.30	1.31	5.84	3.65	0.28
289								+0.125	1.51	1.43	2.94	-0.75	-0.75		2.26	2.22	1.30	1.31	6.24	3.64	0.27
290	98.00	2.00	20.0	23.0	36	55	-0.417	-0.269	1.60	1.43	3.03*	-1.21	-0.71	1.53+1.9	2.32	2.41	1.29	1.30	4.89	3.54	0.29
291								-0.169	1.59	1.43	3.02*	-1.05	-0.80		2.36	2.38	1.28	1.30	5.34	3.55	0.29
292								-0.069	1.59	1.43	3.02*	-0.90	-0.90		2.38	2.34	1.28	1.30	5.67	3.54	0.28
293	98.00	2.00	20.0	23.0	37	53	+0.115	-0.083	1.52	1.43	2.95	-0.97	-0.59	1.43-4.5	2.25	2.30	1.32	1.33	5.46	3.65	0.28
294								+0.017	1.51	1.43	2.94	-0.85	-0.66		2.27	2.28	1.32	1.33	5.94	3.67	0.28
295								+0.117	1.51	1.43	2.94	-0.74	-0.74		2.28	2.26	1.31	1.33	6.38	3.66	0.27
296	98.00	2.00	20.0	23.0	37	54	-0.417	-0.285	1.60	1.43	3.03*	-1.19	-0.70	1.46-2.7	2.34	2.45	1.31	1.32	4.99	3.56	0.29
297								-0.185	1.59	1.43	3.02*	-1.03	-0.79		2.38	2.42	1.30	1.32	5.44	3.57	0.29

298								-0.085	1.59	1.43	3.02*	-0.89-0.89		2.41	2.39	1.30	1.32	5.80	3.57	0.29
299	99.00	2.00	20.0	23.0	35	55	+0.638	+0.227	1.43	1.43	2.86	-0.74-0.57	1.57+4.8	2.12	2.10	1.29	1.30	6.22	3.71	0.27
300								+0.327	1.42	1.43	2.85	-0.64-0.64		2.13	2.08	1.28	1.30	6.67	3.71	0.27
301	99.00	2.00	20.0	23.0	36	54	+0.638	+0.226	1.43	1.43	2.86	-0.73-0.57	1.50+0.0	2.16	2.14	1.31	1.32	6.34	3.73	0.27
302								+0.326	1.43	1.43	2.86	-0.64-0.64		2.17	2.12	1.30	1.32	6.82	3.73	0.27
303	99.00	2.00	20.0	23.0	36	55	+0.071	-0.089	1.52	1.43	2.95	-1.00-0.60	1.53+1.9	2.23	2.27	1.31	1.32	5.37	3.64	0.28
304								+0.011	1.52	1.43	2.95	-0.87-0.67		2.25	2.25	1.30	1.32	5.85	3.65	0.28
305								+0.111	1.51	1.43	2.94	-0.75-0.75		2.27	2.23	1.30	1.32	6.24	3.65	0.28
306	99.00	2.00	20.0	23.0	36	56	-0.458	-0.279	1.60	1.43	3.03*	-1.22-0.71	1.56+3.7	2.33	2.42	1.29	1.31	4.91	3.55	0.29
307								-0.179	1.60	1.43	3.03*	-1.05-0.81		2.37	2.39	1.29	1.31	5.36	3.55	0.29
308								-0.079	1.59	1.43	3.02*	-0.90-0.90		2.39	2.35	1.28	1.30	5.68	3.55	0.29
309	99.00	2.00	20.0	23.0	37	53	+0.638	+0.225	1.43	1.43	2.86	-0.72-0.56	1.43-4.5	2.18	2.18	1.32	1.34	6.46	3.75	0.28
310								+0.325	1.43	1.43	2.86	-0.63-0.63		2.19	2.16	1.32	1.34	6.97	3.75	0.27
311	99.00	2.00	20.0	23.0	37	54	+0.071	-0.097	1.52	1.43	2.95	-0.98-0.59	1.46-2.7	2.26	2.31	1.32	1.34	5.47	3.66	0.28
312								+0.003	1.52	1.43	2.95	-0.86-0.67		2.28	2.29	1.32	1.34	5.96	3.67	0.28
313								+0.103	1.52	1.43	2.95	-0.75-0.75		2.29	2.27	1.32	1.33	6.39	3.67	0.28
314	99.00	2.00	20.0	23.0	37	55	-0.458	-0.295	1.60	1.43	3.03*	-1.20-0.70	1.49-0.9	2.35	2.47	1.31	1.33	5.01	3.57	0.29
315								-0.195	1.60	1.43	3.03*	-1.04-0.80		2.39	2.44	1.31	1.32	5.46	3.57	0.29
316								-0.095	1.59	1.43	3.02*	-0.89-0.89		2.42	2.40	1.30	1.32	5.81	3.57	0.29
317	100.00	2.00	20.0	23.0	36	55	+0.591	+0.108	1.44	1.43	2.87	-0.83-0.51	1.53+1.9	2.16	2.16	1.31	1.33	5.83	3.72	0.28
318								+0.208	1.44	1.43	2.87	-0.73-0.57		2.17	2.14	1.31	1.33	6.36	3.74	0.28
319								+0.308	1.43	1.43	2.86	-0.64-0.64		2.18	2.13	1.31	1.33	6.83	3.74	0.27
320	100.00	2.00	20.0	23.0	36	56	+0.027	-0.102	1.53	1.43	2.96	-1.00-0.60	1.56+3.7	2.24	2.28	1.31	1.33	5.38	3.65	0.28
321								-0.002	1.53	1.43	2.96	-0.88-0.68		2.26	2.26	1.31	1.32	5.87	3.66	0.28
322								+0.098	1.52	1.43	2.95	-0.76-0.76		2.28	2.24	1.30	1.32	6.25	3.66	0.28
323	100.00	2.00	20.0	23.0	37	54	+0.591	+0.106	1.44	1.43	2.87	-0.82-0.50	1.46-2.7	2.18	2.20	1.33	1.35	5.94	3.74	0.28
324								+0.206	1.44	1.43	2.87	-0.72-0.57		2.19	2.18	1.33	1.34	6.48	3.76	0.28
325								+0.306	1.44	1.43	2.87	-0.63-0.63		2.20	2.17	1.33	1.34	6.98	3.75	0.27
326	100.00	2.00	20.0	23.0	37	55	+0.027	-0.111	1.53	1.43	2.96	-0.99-0.59	1.49-0.9	2.26	2.32	1.33	1.34	5.49	3.67	0.28
327								-0.011	1.53	1.43	2.96	-0.86-0.67		2.29	2.30	1.32	1.34	5.98	3.68	0.28
328								+0.089	1.52	1.43	2.95	-0.75-0.75		2.30	2.28	1.32	1.34	6.39	3.68	0.28
329	100.00	2.00	20.0	23.0	37	56	-0.498	-0.304	1.61	1.43	3.04*	-1.21-0.71	1.51+0.9	2.36	2.48	1.31	1.33	5.02	3.58	0.29
330								-0.204	1.61	1.43	3.04*	-1.04-0.80		2.40	2.45	1.31	1.33	5.48	3.58	0.29
331								-0.104	1.60	1.43	3.03*	-0.90-0.90		2.43	2.41	1.31	1.32	5.82	3.58	0.29
332	100.00	2.00	20.0	23.0	38	55	-0.498	-0.320	1.61	1.43	3.04*	-1.19-0.70	1.45-3.5	2.40	2.52	1.33	1.35	5.12	3.59	0.29

333	-0.220 1.61 1.43 3.04*-1.03-0.79	2.44	2.49	1.33	1.34	5.58	3.60	0.29
334	-0.120 1.60 1.43 3.03*-0.89-0.89	2.47	2.45	1.32	1.34	5.95	3.60	0.29

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